# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 79.2 ± 2.96 |
| Correlation Stability | 72.6 ± 3.93 |
| Distribution Consistency | 81.1 ± 6.07 |
| Entropy Preservation | 90.9 ± 1.51 |
| Mean Z Score Drift | 0.1 ± 0.04 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 24.3 ± 0.90 |
| Data Density | 100.0 ± 0.02 |
| Coverage Score | 72.4 ± 2.65 |
| Feature Accuracy | 88.0 ± 3.69 |
| Novelty Score | 0.6 ± 0.02 |

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## Dataset Audit: heart_disease.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 303 |
| Columns | 14 |
| Numeric features | 14 |
| Categorical features | 0 |
| Missing cells (%) | 0.1% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4.1 |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| age | 54.439 | 56.942 | +2.503 | 4.6% | OK |
| sex | 0.680 | 0.583 | -0.097 | 14.3% | Warn |
| cp | 3.158 | 3.037 | -0.121 | 3.8% | OK |
| trestbps | 131.690 | 135.905 | +4.215 | 3.2% | OK |
| chol | 246.693 | 260.851 | +14.158 | 5.7% | OK |
| fbs | 0.149 | 0.149 | +0.000 | 0.2% | OK |
| restecg | 0.990 | 1.256 | +0.266 | 26.9% | Drift |
| thalach | 149.607 | 149.826 | +0.219 | 0.1% | OK |
| exang | 0.327 | 0.455 | +0.128 | 39.1% | Drift |
| oldpeak | 1.040 | 1.138 | +0.099 | 9.5% | OK |
| slope | 1.601 | 1.562 | -0.039 | 2.4% | OK |
| ca | 0.672 | 0.785 | +0.113 | 16.8% | Warn |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 79.2 ± 2.96 |
| Correlation Stability (*2) | 72.6 ± 3.93 |
| Distribution Consistency (*3) | 81.1 ± 6.07 |
| Entropy Preservation (*4) | 90.9 ± 1.51 |
| Mean Z Score Drift (*5) | 0.1 ± 0.04 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 24.3 ± 0.90 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.02 |
| Coverage Score (*9) | 72.4 ± 2.65 |
| Feature Accuracy (*10) | 88.0 ± 3.69 |
| Novelty Score (*11) | 0.6 ± 0.02 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5540983606557377 | 0.4672131147540983 | 0.5098360655737706 | -0.04426229508196724 | |
| F1 | 0.4442202071436568 4 | 0.4422216313950405 6 | 0.4504353226181503 | 0.006215115474493415 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5245901639344263 | 0.5016393442622951 | 0.5065573770491804 | -0.0180327868852459 | |
| F1 | 0.5277823248154705 | 0.5186268185086629 | 0.5149910025974769 | -0.012791322217993607 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5360655737704919 | 0.519672131147541 | 0.5442622950819672 | 0.008196721311475391 | |
| F1 | 0.5239813068292687 | 0.5246752770384715 | 0.5352280337197531 | 0.011246726890484476 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5475409836065573 | 0.4672131147540984 4 | 0.5442622950819672 | -0.00327868852459015 66 | |
| F1 | 0.3899270600110827 | 0.4061770145533976 | 0.3995643691936795 | 0.00963730918259682 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5983606557377049 | 0.4606557377049197 | 0.5868852459016394 | -0.01147540983606557 | |
| F1 | 0.543296686559312 | 0.4677096400641685 | 0.5615921097713189 | 0.01829542321200679 | |

### Model: LR

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| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5983606557377049 | 0.5278688524590163 | 0.5590163934426229 | -0.03934426229508199 | |
| F1 | 0.5650748554025102 | 0.529220649993778 | 0.5443427243979047 | -0.02073213100460557 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5426229508196723 | 0.5262295081967212 | 0.5278688524590164 | -0.01475409836065749 | |
| F1 | 0.5198013329305267 | 0.523751939432127 | 0.5236060588452152 | 0.00380472591468841 7 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5983606557377048 | 0.5426229508196722 | 0.5655737704918032 | -0.03278688524590165 | |
| F1 | 0.5545478162546738 | 0.533337127955305 | 0.5362177244791138 | -0.01833009177555990 4 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6081967213114754 | 0.4229508196721311 | 0.5885245901639344 | -0.01967213114754098 2 | |
| F1 | 0.6070236204175147 | 0.43286480864235416 | 0.5857404433903651 | -0.02128317702714946 4 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5573770491803279 | 0.4737704918032787 4 | 0.5295081967213114 | -0.02786885245901636 3 | |
| F1 | 0.568660979990255 | 0.4959247876547034 4 | 0.5484799031518419 | -0.0201810768384132 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 63.0 | 1.0 | 1.0 | 145.0 | 233.0 | 1.0 | 2.0 | 150.0 | 0.0 | 2.3 |
| 2 | 67.0 | 1.0 | 4.0 | 160.0 | 286.0 | 0.0 | 2.0 | 108.0 | 1.0 | 1.5 |
| 3 | 67.0 | 1.0 | 4.0 | 120.0 | 229.0 | 0.0 | 2.0 | 129.0 | 1.0 | 2.6 |
| 4 | 37.0 | 1.0 | 3.0 | 130.0 | 250.0 | 0.0 | 0.0 | 187.0 | 0.0 | 3.5 |
| 5 | 41.0 | 0.0 | 2.0 | 130.0 | 204.0 | 0.0 | 2.0 | 172.0 | 0.0 | 1.4 |
| 6 | 56.0 | 1.0 | 2.0 | 120.0 | 236.0 | 0.0 | 0.0 | 178.0 | 0.0 | 0.8 |
| 7 | 62.0 | 0.0 | 4.0 | 140.0 | 268.0 | 0.0 | 2.0 | 160.0 | 0.0 | 3.6 |
| 8 | 57.0 | 0.0 | 4.0 | 120.0 | 354.0 | 0.0 | 0.0 | 163.0 | 1.0 | 0.6 |
| 9 | 63.0 | 1.0 | 4.0 | 130.0 | 254.0 | 0.0 | 2.0 | 147.0 | 0.0 | 1.4 |
| 10 | 53.0 | 1.0 | 4.0 | 140.0 | 203.0 | 1.0 | 2.0 | 155.0 | 1.0 | 3.1 |
| 11 | 57.0 | 1.0 | 4.0 | 140.0 | 192.0 | 0.0 | 0.0 | 148.0 | 0.0 | 0.4 |
| 12 | 56.0 | 0.0 | 2.0 | 140.0 | 294.0 | 0.0 | 2.0 | 153.0 | 0.0 | 1.3 |
| 13 | 56.0 | 1.0 | 3.0 | 130.0 | 256.0 | 1.0 | 2.0 | 142.0 | 1.0 | 0.6 |
| 14 | 44.0 | 1.0 | 2.0 | 120.0 | 263.0 | 0.0 | 0.0 | 173.0 | 0.0 | 0.0 |
| 15 | 52.0 | 1.0 | 3.0 | 172.0 | 199.0 | 1.0 | 0.0 | 162.0 | 0.0 | 0.5 |
| 16 | 57.0 | 1.0 | 3.0 | 150.0 | 168.0 | 0.0 | 0.0 | 174.0 | 0.0 | 1.6 |
| 17 | 48.0 | 1.0 | 2.0 | 110.0 | 229.0 | 0.0 | 0.0 | 168.0 | 0.0 | 1.0 |
| 18 | 54.0 | 1.0 | 4.0 | 140.0 | 239.0 | 0.0 | 0.0 | 160.0 | 0.0 | 1.2 |
| 19 | 48.0 | 0.0 | 3.0 | 130.0 | 275.0 | 0.0 | 0.0 | 139.0 | 0.0 | 0.2 |
| 20 | 49.0 | 1.0 | 2.0 | 130.0 | 266.0 | 0.0 | 0.0 | 171.0 | 0.0 | 0.6 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 61.0 | 0.0 | 4.0 | 130.0 | 330.0 | 0.0 | 2.0 | 169.0 | 0.0 | 0.0 |
| 2 | 41.0 | 1.0 | 3.0 | 112.0 | 250.0 | 0.0 | 0.0 | 179.0 | 0.0 | 0.0 |
| 3 | 43.0 | 1.0 | 4.0 | 115.0 | 303.0 | 0.0 | 0.0 | 181.0 | 0.0 | 1.2 |
| 4 | 55.0 | 1.0 | 4.0 | 140.0 | 217.0 | 0.0 | 0.0 | 111.0 | 1.0 | 5.6 |
| 5 | 70.0 | 1.0 | 4.0 | 145.0 | 174.0 | 0.0 | 0.0 | 125.0 | 1.0 | 2.6 |
| 6 | 61.0 | 1.0 | 4.0 | 148.0 | 203.0 | 0.0 | 0.0 | 161.0 | 0.0 | 0.0 |
| 7 | 48.0 | 1.0 | 3.0 | 138.0 | 256.0 | 0.0 | 2.0 | 172.0 | 0.0 | 0.7 |
| 8 | 62.0 | 0.0 | 4.0 | 142.0 | 310.0 | 0.0 | 0.0 | 134.0 | 1.0 | 1.5 |
| 9 | 59.0 | 1.0 | 2.0 | 128.0 | 275.0 | 1.0 | 0.0 | 164.0 | 0.0 | 0.0 |
| 10 | 53.0 | 0.0 | 3.0 | 135.0 | 245.0 | 0.0 | 2.0 | 148.0 | 0.0 | 0.4 |
| 11 | 66.0 | 1.0 | 4.0 | 150.0 | 289.0 | 0.0 | 2.0 | 120.0 | 1.0 | 2.7 |
| 12 | 57.0 | 1.0 | 2.0 | 124.0 | 231.0 | 0.0 | 0.0 | 158.0 | 0.0 | 0.0 |
| 13 | 49.0 | 0.0 | 4.0 | 130.0 | 269.0 | 0.0 | 0.0 | 163.0 | 0.0 | 0.0 |
| 14 | 54.0 | 1.0 | 3.0 | 120.0 | 258.0 | 0.0 | 2.0 | 147.0 | 0.0 | 0.4 |
| 15 | 55.0 | 1.0 | 2.0 | 130.0 | 262.0 | 0.0 | 0.0 | 155.0 | 0.0 | 0.0 |
| 16 | 46.0 | 1.0 | 2.0 | 101.0 | 197.0 | 1.0 | 0.0 | 156.0 | 0.0 | 0.0 |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 54.0 | 1.0 | 4.0 | 122.0 | 286.0 | 0.0 | 2.0 | 116.0 | 1.0 | 3.2 |
| 2 | 62.0 | 1.0 | 2.0 | 128.0 | 208.0 | 1.0 | 2.0 | 140.0 | 0.0 | 0.0 |
| 3 | 67.0 | 0.0 | 4.0 | 106.0 | 223.0 | 0.0 | 0.0 | 142.0 | 0.0 | 0.3 |
| 4 | 38.0 | 1.0 | 3.0 | 138.0 | 175.0 | 0.0 | 0.0 | 173.0 | 0.0 | 0.0 |
| 5 | 44.0 | 1.0 | 2.0 | 130.0 | 219.0 | 0.0 | 2.0 | 188.0 | 0.0 | 0.0 |
| 6 | 59.0 | 1.0 | 4.0 | 140.0 | 177.0 | 0.0 | 0.0 | 162.0 | 1.0 | 0.0 |
| 7 | 61.0 | 1.0 | 3.0 | 134.0 | 250.0 | 0.0 | 2.0 | 145.0 | 0.0 | 1.2 |
| 8 | 48.0 | 0.0 | 2.0 | 128.0 | 210.0 | 0.0 | 0.0 | 170.0 | 0.0 | 0.0 |
| 9 | 55.0 | 1.0 | 4.0 | 142.0 | 289.0 | 1.0 | 2.0 | 120.0 | 1.0 | 2.8 |
| 10 | 63.0 | 1.0 | 3.0 | 138.0 | 230.0 | 0.0 | 2.0 | 150.0 | 0.0 | 0.6 |
| 11 | 46.0 | 1.0 | 2.0 | 135.0 | 195.0 | 0.0 | 0.0 | 180.0 | 0.0 | 0.0 |
| 12 | 69.0 | 0.0 | 4.0 | 158.0 | 310.0 | 1.0 | 2.0 | 128.0 | 1.0 | 3.5 |
| 13 | 63.0 | 0.0 | 3.0 | 135.0 | 252.0 | 0.0 | 2.0 | 172.0 | 0.0 | 0.0 |
| 14 | 60.0 | 1.0 | 4.0 | 130.0 | 253.0 | 0.0 | 0.0 | 144.0 | 1.0 | 1.4 |
| 15 | 58.0 | 0.0 | 2.0 | 136.0 | 319.0 | 1.0 | 2.0 | 152.0 | 0.0 | 0.0 |
| 16 | 59.0 | 0.0 | 4.0 | 174.0 | 249.0 | 0.0 | 0.0 | 143.0 | 1.0 | 0.0 |

## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 40.0 | 1.0 | 4.0 | 110.0 | 167.0 | 0.0 | 2.0 | 114.0 | 1.0 | 2.0 |
| 2 | 37.0 | 0.0 | 3.0 | 120.0 | 215.0 | 0.0 | 0.0 | 170.0 | 0.0 | 0.0 |
| 3 | 55.0 | 0.0 | 4.0 | 180.0 | 327.0 | 0.0 | 1.0 | 117.0 | 1.0 | 3.4 |
| 4 | 43.0 | 1.0 | 4.0 | 115.0 | 303.0 | 0.0 | 0.0 | 181.0 | 0.0 | 1.2 |
| 5 | 65.0 | 0.0 | 3.0 | 160.0 | 360.0 | 0.0 | 2.0 | 151.0 | 0.0 | 0.8 |
| 6 | 63.0 | 0.0 | 4.0 | 124.0 | 197.0 | 0.0 | 0.0 | 136.0 | 1.0 | 0.0 |
| 7 | 57.0 | 1.0 | 3.0 | 128.0 | 245.0 | 0.0 | 2.0 | 142.0 | 1.0 | 2.4 |
| 8 | 48.0 | 0.0 | 2.0 | 124.0 | 210.0 | 0.0 | 0.0 | 168.0 | 0.0 | 0.2 |
| 9 | 60.0 | 1.0 | 4.0 | 138.0 | 289.0 | 0.0 | 2.0 | 112.0 | 1.0 | 3.0 |
| 10 | 53.0 | 1.0 | 1.0 | 142.0 | 260.0 | 0.0 | 0.0 | 155.0 | 0.0 | 1.0 |
| 11 | 45.0 | 0.0 | 3.0 | 118.0 | 225.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 12 | 66.0 | 1.0 | 4.0 | 152.0 | 310.0 | 0.0 | 2.0 | 120.0 | 1.0 | 2.8 |
| 13 | 49.0 | 1.0 | 3.0 | 120.0 | 188.0 | 0.0 | 0.0 | 139.0 | 0.0 | 2.0 |
| 14 | 45.0 | 1.0 | 4.0 | 115.0 | 260.0 | 0.0 | 2.0 | 185.0 | 0.0 | 0.0 |
| 15 | 45.0 | 0.0 | 2.0 | 130.0 | 234.0 | 0.0 | 2.0 | 175.0 | 0.0 | 0.6 |
| 16 | 46.0 | 1.0 | 3.0 | 150.0 | 231.0 | 0.0 | 0.0 | 147.0 | 0.0 | 3.6 |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 56.0 | 1.0 | 2.0 | 120.0 | 240.0 | 0.0 | 0.0 | 169.0 | 0.0 | 0.0 |
| 2 | 68.0 | 1.0 | 3.0 | 118.0 | 277.0 | 0.0 | 0.0 | 151.0 | 0.0 | 1.0 |
| 3 | 43.0 | 1.0 | 4.0 | 120.0 | 177.0 | 0.0 | 2.0 | 120.0 | 1.0 | 2.5 |
| 4 | 68.0 | 1.0 | 4.0 | 144.0 | 193.0 | 1.0 | 0.0 | 141.0 | 0.0 | 3.4 |
| 5 | 60.0 | 1.0 | 3.0 | 140.0 | 185.0 | 0.0 | 2.0 | 155.0 | 0.0 | 3.0 |
| 6 | 48.0 | 1.0 | 4.0 | 122.0 | 222.0 | 0.0 | 2.0 | 186.0 | 0.0 | 0.0 |
| 7 | 61.0 | 1.0 | 4.0 | 142.0 | 260.0 | 0.0 | 2.0 | 138.0 | 1.0 | 2.7 |
| 8 | 46.0 | 0.0 | 3.0 | 124.0 | 210.0 | 0.0 | 0.0 | 170.0 | 0.0 | 0.3 |
| 9 | 53.0 | 1.0 | 2.0 | 135.0 | 250.0 | 1.0 | 2.0 | 158.0 | 0.0 | 1.1 |
| 10 | 67.0 | 1.0 | 4.0 | 148.0 | 234.0 | 0.0 | 0.0 | 120.0 | 1.0 | 3.0 |
| 11 | 55.0 | 0.0 | 3.0 | 128.0 | 192.0 | 0.0 | 2.0 | 165.0 | 0.0 | 0.0 |
| 12 | 50.0 | 1.0 | 1.0 | 138.0 | 289.0 | 0.0 | 2.0 | 142.0 | 1.0 | 1.4 |
| 13 | 57.0 | 1.0 | 4.0 | 140.0 | 192.0 | 0.0 | 0.0 | 148.0 | 0.0 | 0.4 |
| 14 | 63.0 | 0.0 | 4.0 | 150.0 | 407.0 | 0.0 | 2.0 | 154.0 | 0.0 | 4.0 |
| 15 | 57.0 | 0.0 | 4.0 | 128.0 | 303.0 | 0.0 | 2.0 | 159.0 | 0.0 | 0.0 |
| 16 | 60.0 | 1.0 | 4.0 | 117.0 | 230.0 | 1.0 | 0.0 | 160.0 | 1.0 | 1.4 |

## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 53.0 | 1.0 | 3.0 | 130.0 | 197.0 | 1.0 | 2.0 | 152.0 | 0.0 | 1.2 |
| 2 | 53.0 | 0.0 | 3.0 | 128.0 | 216.0 | 0.0 | 2.0 | 115.0 | 0.0 | 0.0 |
| 3 | 63.0 | 0.0 | 3.0 | 135.0 | 252.0 | 0.0 | 2.0 | 172.0 | 0.0 | 0.0 |
| 4 | 57.0 | 1.0 | 2.0 | 124.0 | 261.0 | 0.0 | 0.0 | 141.0 | 0.0 | 0.3 |
| 5 | 46.0 | 1.0 | 4.0 | 120.0 | 249.0 | 0.0 | 2.0 | 144.0 | 0.0 | 0.8 |
| 6 | 62.0 | 1.0 | 2.0 | 128.0 | 208.0 | 1.0 | 2.0 | 140.0 | 0.0 | 0.0 |
| 7 | 61.0 | 1.0 | 4.0 | 142.0 | 260.0 | 0.0 | 2.0 | 138.0 | 1.0 | 2.6 |
| 8 | 48.0 | 0.0 | 3.0 | 130.0 | 220.0 | 0.0 | 0.0 | 170.0 | 0.0 | 0.0 |
| 9 | 59.0 | 1.0 | 2.0 | 128.0 | 245.0 | 1.0 | 2.0 | 154.0 | 0.0 | 1.1 |
| 10 | 56.0 | 1.0 | 4.0 | 135.0 | 298.0 | 0.0 | 2.0 | 120.0 | 1.0 | 2.3 |
| 11 | 52.0 | 1.0 | 3.0 | 120.0 | 210.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 12 | 65.0 | 0.0 | 4.0 | 150.0 | 312.0 | 0.0 | 2.0 | 108.0 | 1.0 | 1.8 |
| 13 | 50.0 | 0.0 | 2.0 | 120.0 | 244.0 | 0.0 | 0.0 | 162.0 | 0.0 | 1.1 |
| 14 | 56.0 | 1.0 | 2.0 | 120.0 | 236.0 | 0.0 | 0.0 | 178.0 | 0.0 | 0.8 |
| 15 | 61.0 | 1.0 | 3.0 | 150.0 | 243.0 | 1.0 | 0.0 | 137.0 | 1.0 | 1.0 |
| 16 | 64.0 | 0.0 | 3.0 | 140.0 | 313.0 | 0.0 | 0.0 | 133.0 | 0.0 | 0.2 |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.0 | 0.0 | 1.0 | 150.0 | 240.0 | 0.0 | 0.0 | 171.0 | 0.0 | 0.9 |
| 2 | 45.0 | 1.0 | 4.0 | 104.0 | 208.0 | 0.0 | 2.0 | 148.0 | 1.0 | 3.0 |
| 3 | 59.0 | 1.0 | 3.0 | 150.0 | 212.0 | 1.0 | 0.0 | 157.0 | 0.0 | 1.6 |
| 4 | 47.0 | 1.0 | 3.0 | 108.0 | 243.0 | 0.0 | 0.0 | 152.0 | 0.0 | 0.0 |
| 5 | 43.0 | 1.0 | 4.0 | 150.0 | 247.0 | 0.0 | 0.0 | 171.0 | 0.0 | 1.5 |
| 6 | 57.0 | 1.0 | 4.0 | 110.0 | 335.0 | 0.0 | 0.0 | 143.0 | 1.0 | 3.0 |
| 7 | 49.0 | 1.0 | 3.0 | 138.0 | 215.0 | 0.0 | 0.0 | 154.0 | 0.0 | 0.8 |
| 8 | 53.0 | 0.0 | 2.0 | 128.0 | 270.0 | 0.0 | 2.0 | 162.0 | 0.0 | 1.0 |
| 9 | 61.0 | 1.0 | 4.0 | 142.0 | 289.0 | 0.0 | 0.0 | 120.0 | 1.0 | 2.4 |
| 10 | 46.0 | 1.0 | 1.0 | 124.0 | 199.0 | 0.0 | 2.0 | 172.0 | 0.0 | 0.0 |
| 11 | 58.0 | 0.0 | 3.0 | 135.0 | 245.0 | 1.0 | 0.0 | 138.0 | 1.0 | 1.6 |
| 12 | 37.0 | 1.0 | 2.0 | 120.0 | 220.0 | 0.0 | 0.0 | 168.0 | 0.0 | 0.0 |
| 13 | 39.0 | 1.0 | 4.0 | 118.0 | 219.0 | 0.0 | 0.0 | 140.0 | 0.0 | 1.2 |
| 14 | 62.0 | 0.0 | 4.0 | 138.0 | 294.0 | 1.0 | 0.0 | 106.0 | 0.0 | 1.9 |
| 15 | 66.0 | 1.0 | 4.0 | 160.0 | 228.0 | 0.0 | 2.0 | 138.0 | 0.0 | 2.3 |
| 16 | 39.0 | 1.0 | 3.0 | 140.0 | 321.0 | 0.0 | 2.0 | 182.0 | 0.0 | 0.0 |

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 65.0 | 0.0 | 3.0 | 155.0 | 269.0 | 0.0 | 0.0 | 148.0 | 0.0 | 0.8 |
| 2 | 71.0 | 0.0 | 4.0 | 112.0 | 149.0 | 0.0 | 0.0 | 125.0 | 0.0 | 1.6 |
| 3 | 61.0 | 1.0 | 1.0 | 134.0 | 234.0 | 0.0 | 0.0 | 145.0 | 0.0 | 2.6 |
| 4 | 64.0 | 1.0 | 3.0 | 140.0 | 335.0 | 0.0 | 0.0 | 158.0 | 0.0 | 0.0 |
| 5 | 55.0 | 1.0 | 2.0 | 130.0 | 262.0 | 0.0 | 0.0 | 155.0 | 0.0 | 0.0 |
| 6 | 48.0 | 1.0 | 2.0 | 110.0 | 229.0 | 0.0 | 0.0 | 168.0 | 0.0 | 1.0 |
| 7 | 58.0 | 1.0 | 3.0 | 128.0 | 236.0 | 0.0 | 0.0 | 155.0 | 0.0 | 0.7 |
| 8 | 63.0 | 0.0 | 4.0 | 142.0 | 289.0 | 0.0 | 2.0 | 134.0 | 1.0 | 2.1 |
| 9 | 52.0 | 1.0 | 2.0 | 138.0 | 210.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 10 | 60.0 | 0.0 | 3.0 | 125.0 | 245.0 | 0.0 | 2.0 | 160.0 | 0.0 | 1.2 |
| 11 | 49.0 | 1.0 | 4.0 | 135.0 | 298.0 | 0.0 | 0.0 | 148.0 | 1.0 | 1.9 |
| 12 | 57.0 | 0.0 | 3.0 | 118.0 | 264.0 | 0.0 | 0.0 | 162.0 | 0.0 | 0.0 |
| 13 | 56.0 | 0.0 | 4.0 | 134.0 | 409.0 | 0.0 | 2.0 | 150.0 | 1.0 | 1.9 |
| 14 | 58.0 | 1.0 | 3.0 | 105.0 | 240.0 | 0.0 | 2.0 | 154.0 | 1.0 | 0.6 |
| 15 | 54.0 | 1.0 | 2.0 | 192.0 | 283.0 | 0.0 | 2.0 | 195.0 | 0.0 | 0.0 |
| 16 | 57.0 | 1.0 | 4.0 | 110.0 | 201.0 | 0.0 | 0.0 | 126.0 | 1.0 | 1.5 |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 41.0 | 0.0 | 2.0 | 130.0 | 204.0 | 0.0 | 2.0 | 172.0 | 0.0 | 1.4 |
| 2 | 39.0 | 0.0 | 3.0 | 94.0 | 199.0 | 0.0 | 0.0 | 179.0 | 0.0 | 0.0 |
| 3 | 61.0 | 0.0 | 4.0 | 130.0 | 330.0 | 0.0 | 2.0 | 169.0 | 0.0 | 0.0 |
| 4 | 67.0 | 1.0 | 4.0 | 160.0 | 286.0 | 0.0 | 2.0 | 108.0 | 1.0 | 1.5 |
| 5 | 60.0 | 1.0 | 4.0 | 130.0 | 253.0 | 0.0 | 0.0 | 144.0 | 1.0 | 1.4 |
| 6 | 58.0 | 1.0 | 3.0 | 140.0 | 211.0 | 1.0 | 2.0 | 165.0 | 0.0 | 0.0 |
| 7 | 56.0 | 1.0 | 3.0 | 128.0 | 236.0 | 0.0 | 2.0 | 158.0 | 1.0 | 0.8 |
| 8 | 48.0 | 0.0 | 2.0 | 110.0 | 210.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 9 | 62.0 | 1.0 | 4.0 | 145.0 | 278.0 | 1.0 | 2.0 | 130.0 | 1.0 | 2.2 |
| 10 | 53.0 | 1.0 | 3.0 | 138.0 | 250.0 | 0.0 | 2.0 | 155.0 | 0.0 | 0.6 |
| 11 | 59.0 | 0.0 | 4.0 | 132.0 | 294.0 | 0.0 | 2.0 | 120.0 | 1.0 | 1.5 |
| 12 | 45.0 | 1.0 | 2.0 | 120.0 | 185.0 | 0.0 | 0.0 | 168.0 | 0.0 | 0.0 |
| 13 | 52.0 | 1.0 | 1.0 | 118.0 | 186.0 | 0.0 | 2.0 | 190.0 | 0.0 | 0.0 |
| 14 | 54.0 | 0.0 | 3.0 | 135.0 | 304.0 | 1.0 | 0.0 | 170.0 | 0.0 | 0.0 |
| 15 | 61.0 | 1.0 | 4.0 | 140.0 | 207.0 | 0.0 | 2.0 | 138.0 | 1.0 | 1.9 |
| 16 | 61.0 | 1.0 | 3.0 | 150.0 | 243.0 | 1.0 | 0.0 | 137.0 | 1.0 | 1.0 |

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 39.0 | 1.0 | 4.0 | 118.0 | 219.0 | 0.0 | 0.0 | 140.0 | 0.0 | 1.2 |
| 2 | 67.0 | 0.0 | 3.0 | 115.0 | 564.0 | 0.0 | 2.0 | 160.0 | 0.0 | 1.6 |
| 3 | 55.0 | 1.0 | 4.0 | 140.0 | 217.0 | 0.0 | 0.0 | 111.0 | 1.0 | 5.6 |
| 4 | 49.0 | 1.0 | 3.0 | 118.0 | 149.0 | 0.0 | 2.0 | 126.0 | 0.0 | 0.8 |
| 5 | 54.0 | 0.0 | 3.0 | 108.0 | 267.0 | 0.0 | 2.0 | 167.0 | 0.0 | 0.0 |
| 6 | 52.0 | 1.0 | 2.0 | 120.0 | 325.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.2 |
| 7 | 50.0 | 1.0 | 3.0 | 128.0 | 245.0 | 0.0 | 2.0 | 148.0 | 0.0 | 0.7 |
| 8 | 61.0 | 0.0 | 4.0 | 135.0 | 260.0 | 1.0 | 2.0 | 140.0 | 0.0 | 1.4 |
| 9 | 57.0 | 1.0 | 2.0 | 142.0 | 230.0 | 0.0 | 0.0 | 165.0 | 0.0 | 0.0 |
| 10 | 46.0 | 1.0 | 4.0 | 120.0 | 210.0 | 0.0 | 2.0 | 170.0 | 0.0 | 0.5 |
| 11 | 65.0 | 1.0 | 3.0 | 150.0 | 240.0 | 0.0 | 2.0 | 128.0 | 1.0 | 2.2 |
| 12 | 53.0 | 0.0 | 2.0 | 122.0 | 198.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 13 | 44.0 | 0.0 | 3.0 | 118.0 | 242.0 | 0.0 | 0.0 | 149.0 | 0.0 | 0.3 |
| 14 | 38.0 | 1.0 | 1.0 | 120.0 | 231.0 | 0.0 | 0.0 | 182.0 | 1.0 | 3.8 |
| 15 | 49.0 | 1.0 | 3.0 | 120.0 | 188.0 | 0.0 | 0.0 | 139.0 | 0.0 | 2.0 |
| 16 | 61.0 | 1.0 | 4.0 | 120.0 | 260.0 | 0.0 | 0.0 | 140.0 | 1.0 | 3.6 |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 39.0 | 0.0 | 3.0 | 94.0 | 199.0 | 0.0 | 0.0 | 179.0 | 0.0 | 0.0 |
| 2 | 44.0 | 1.0 | 4.0 | 110.0 | 197.0 | 0.0 | 2.0 | 177.0 | 0.0 | 0.0 |
| 3 | 56.0 | 1.0 | 2.0 | 130.0 | 221.0 | 0.0 | 2.0 | 163.0 | 0.0 | 0.0 |
| 4 | 65.0 | 1.0 | 4.0 | 120.0 | 177.0 | 0.0 | 0.0 | 140.0 | 0.0 | 0.4 |
| 5 | 55.0 | 1.0 | 4.0 | 160.0 | 289.0 | 0.0 | 2.0 | 145.0 | 1.0 | 0.8 |
| 6 | 47.0 | 1.0 | 3.0 | 130.0 | 253.0 | 0.0 | 0.0 | 179.0 | 0.0 | 0.0 |
| 7 | 53.0 | 1.0 | 4.0 | 128.0 | 245.0 | 0.0 | 2.0 | 143.0 | 1.0 | 2.3 |
| 8 | 46.0 | 0.0 | 3.0 | 118.0 | 210.0 | 0.0 | 0.0 | 172.0 | 0.0 | 0.0 |
| 9 | 62.0 | 1.0 | 2.0 | 138.0 | 294.0 | 0.0 | 2.0 | 155.0 | 0.0 | 1.4 |
| 10 | 49.0 | 1.0 | 4.0 | 140.0 | 230.0 | 1.0 | 0.0 | 170.0 | 0.0 | 0.6 |
| 11 | 58.0 | 1.0 | 3.0 | 124.0 | 276.0 | 0.0 | 2.0 | 160.0 | 1.0 | 1.8 |
| 12 | 63.0 | 0.0 | 4.0 | 132.0 | 280.0 | 0.0 | 0.0 | 148.0 | 0.0 | 0.0 |
| 13 | 63.0 | 0.0 | 3.0 | 135.0 | 252.0 | 0.0 | 2.0 | 172.0 | 0.0 | 0.0 |
| 14 | 63.0 | 1.0 | 1.0 | 145.0 | 233.0 | 1.0 | 2.0 | 150.0 | 0.0 | 2.3 |
| 15 | 60.0 | 1.0 | 4.0 | 125.0 | 258.0 | 0.0 | 2.0 | 141.0 | 1.0 | 2.8 |
| 16 | 62.0 | 0.0 | 4.0 | 140.0 | 268.0 | 0.0 | 2.0 | 160.0 | 0.0 | 3.6 |